

Finding the Way

Navigation and Obstacle Avoidance for the Blind and Visually Impaired

User Guide

Circuit Syndicate

The Citadel — Department of Electrical and Computer Engineering

Sponsored by ACMIC INC

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Important Safety Information

Read all safety information before using Finding the Way for the first time.

⚠ WARNING. Finding the Way is a navigation aid and is not a replacement for the white cane, guide dog, or other primary navigation tools. Always use a primary navigation tool, especially in unfamiliar or high-risk environments.

⚠ WARNING. The ultrasonic subsystem covers only the 180° arc directly in front of the chest pack. Objects approaching from the rear are not detected. Stay alert to your full surroundings at all times.

⚠ WARNING. Do not leave the chest-pack lithium-ion battery pack charging unattended or overnight. Dispose of lithium-ion batteries through a certified recycling program, never in household waste.

⚠ WARNING. Do not expose the chest pack, smart glasses, or smartphone to rain, splashing water, or excessive humidity. The current prototype housings carry no formal IP rating.

⚠ WARNING. If the chest-pack battery falls to about 25%, the application will announce a low-battery warning; at about 10% the warning escalates; at about 5% it becomes critical. Recharge the chest pack before the next session. Operating below the critical threshold may cause an unexpected shutdown.

NOTE. The system is designed so that you cannot disable both sensing modalities at once while the application is connected. If the vest detects that the smart glasses are off when you ask to disable ultrasonics, it will keep the vest active and announce a safety override.

What Is in the Box

Your Finding the Way kit includes the components below. Verify all items are present before first use.

Item	Qty	Description
Chest pack vest	1	Adjustable chest-worn vest with four integrated ultrasonic sensors and four haptic vibration motors.
Control housing unit	1	ESP32 control board housed inside the chest pack. Do not remove.
Li-Ion battery pack	1	USB-C rechargeable battery pack (10,000 mAh), pre-installed in the chest pack.
USB-C charging cable	1	For charging the chest-pack battery pack.
Smart glasses	1	OMI Smart Glasses module with Seeed XIAO ESP32-S3 Sense and OV2640 camera. Internal LiPo battery (dual 250 mAh, 500 mAh total).
Smart-glasses charging cable	1	USB-C cable for charging the smart glasses.
Android smartphone	1	User-supplied. Must run Android 7.0 (API 24) or higher and support Bluetooth Low Energy 4.2 or newer.
Finding the Way app (APK)	1	Pre-installed on a provided phone, or installable from the APK.

NOTE. The Android smartphone is user-supplied and is not included in the kit. The application's minimum supported Android version is 7.0 (Nougat, API 24); the current target SDK is 36.

Part 1: Chest Pack

The chest pack is the non-visual sensing component of the system. It continuously samples the frontal environment with four ultrasonic sensors and signals obstacle proximity through four haptic motors distributed across the vest. Each sensor drives one motor, giving you six left-to-right detection zones across the front arc using the overlaps between zones.

1.1 Getting to Know Your Chest Pack

Before use, become familiar with these physical features:

- **Ultrasonic sensors (four):** across the front of the vest. Keep them clear of clothing or accessories that could block their line of sight.
- **Haptic motors (four):** distributed across the vest surface. Each motor corresponds to one ultrasonic sensor zone.
- **Switch 1 (Haptics):** on the right side of the housing unit. Toggles haptic feedback on or off.
- **Switch 2 (Ultrasonics):** on the left side of the housing unit. Toggles ultrasonic sensing on or off. While the application is connected, the safety interlock will prevent disabling ultrasonics if the smart glasses are not active.
- **USB-C port:** used for charging the battery pack.
- **Status LED:** on the front of the housing unit. In normal operation it is illuminated whenever the housing receives power.

NOTE. The two switches on the housing are sampled continuously by the firmware. While the application is not connected, the switch positions are authoritative. While the application is connected, app commands take precedence and the switch state is reported back as a diagnostic mismatch if it differs.

1.2 Putting On the Chest Pack

1. Hold the vest with the sensor array facing outward (away from your body).
2. Slip your arms through the shoulder straps as you would a standard vest.
3. Fasten the front clasp or zipper. The chest pack should sit flat against your sternum with the sensors unobstructed.
4. Adjust the shoulder and side straps until the vest sits comfortably and does not shift during movement. The sensor array should remain level when you stand upright.
5. Make sure no clothing covers the sensor openings. Loose scarves, coat lapels, or front-worn bags can block the sensor beams and reduce detection accuracy.

TIP. If the vest feels loose during walking, tighten the side straps slightly. The haptic motors transmit best when the vest maintains consistent contact with your torso.

1.3 Powering On

1. Make sure the Li-Ion battery pack is charged (see Section 1.6). The pack is pre-installed inside the chest pack.
2. Press the red push button on the inside of the chest pack once. The ultrasonics and haptics will turn on and immediately begin sensing surroundings.

3. The control housing will boot and the BLE radio will start advertising. The chest pack appears as “Navigation Vest” to nearby Bluetooth scanners.
4. Open the Finding the Way application on your smartphone and connect to the chest pack (see Part 3).

1.4 Understanding Haptic Feedback

The haptic system communicates two pieces of information: which zone an obstacle is in and how close it is. The vibration pattern speeds up as the object approaches. Each sensor reading is averaged across a five-sample window and rounded to a whole foot, then mapped to a tier:

Distance (rounded)	Pulse pattern (ON / OFF)	Urgency
2 ft	100 ms / 100 ms (rapid pulse)	Maximum
3 ft	100 ms / 300 ms	Very high
4 ft	100 ms / 500 ms	High
5 ft	100 ms / 700 ms	Moderate
6 ft	100 ms / 900 ms (slow pulse)	Low
Other / no echo	No vibration	None

The location of the vibrating motor on your chest indicates which of the four sensor zones detected an obstacle. Left-side motors signal obstacles to your left; right-side motors signal obstacles to your right; center motors signal obstacles directly ahead.

TIP. When learning the system, practice in a familiar open space. Walk slowly and focus on identifying which motor vibrates and at what rate. 15 to 30 minutes of practice is enough for haptic-zone identification to become intuitive.

1.5 Powering Off

1. The sensors and haptics can be disabled separately while maintaining BLE connection to the smartphone using the Mobile Application.
2. Press the red push button on the inside of the chest pack once. Sensors, Haptics, and BLE will all be turned off.
3. Remove the vest by unfastening the front clasp and slipping your arms out of the shoulder straps.
4. Store the chest pack in a cool, dry location away from direct sunlight.

1.6 Charging the Chest-Pack Battery

1. Remove the pre-connected USB-C charging cable from the inside of the chest pack.
2. Connect the USB-C cable to a wall adapter or other USB-C capable charger rated at 5 V / 2 A or higher.
3. A second, blue LED will energize inside the Chest Pack indicating the charging connection is sound.

4. A full charge from empty takes approximately 4 to 6 hours. Battery life under normal combined-system load is greater than 4 hours.

⚠ WARNING. Do not charge the battery pack while wearing the vest. Do not use a charger rated above 5 V / 3 A. Do not leave the battery charging unattended for extended periods.

1.7 Caring for the Chest Pack

- Wipe the outer surface with a dry or lightly damp cloth. Do not use abrasive cleaners or solvents, and do not spray any cleaning product directly onto the vest.
- Do not submerge in water or wear in heavy rain. The current prototype is not waterproofed.
- Do not fold or compress the chest area of the vest when storing, as this can damage the sensor brackets or wiring.
- Inspect the ultrasonic sensor faces periodically for dust or debris. Clean gently with a dry cotton swab if any buildup is observed.
- Do not open the housing unit or modify any internal components. Doing so could destroy delicate internal connections, making the chest pack unusable
- If a haptic motor stops responding, note which zone is affected and contact support. Do not attempt to replace motors yourself.

Part 2: Smart Glasses

The smart glasses provide the visual object-detection capability of the system. The onboard camera periodically captures images of your environment and transmits them over BLE to your smartphone, where an ONNX Runtime YOLO model identifies objects, estimates distances, and classifies motion before announcing the results aloud through your phone's speaker.

2.1 Getting to Know the Smart Glasses

- **Camera (OV2640):** on the front of the glasses frame. Faces forward when worn normally. Do not obstruct or cover the lens.
- **ESP32-S3 module (Seeed XIAO ESP32-S3 Sense):** integrated into the frame. Houses the processor, BLE radio, and PSRAM buffer. Do not attempt to remove or modify.
- **Internal LiPo battery:** built into the frame (dual 250 mAh, 500 mAh total). Recharged via the glasses' USB-C port.
- **USB-C charging port:** on the underside or side of the frame.
- **Power / Capture button:** a small tactile button on the frame. While powered on and BLE-connected, a short press triggers a priority burst capture (four frames). A long press of about 2 seconds places the glasses into deep sleep.

2.2 Putting On the Smart Glasses

1. Hold the glasses with the camera lens facing outward.
2. Place the frame over your ears as you would standard eyeglasses. The camera should point straight ahead at approximately eye level.
3. Adjust the frame so it sits comfortably without sliding down your nose.
4. If you wear prescription glasses, the smart glasses can be worn over them, though fit may vary.

TIP. Camera alignment matters. If the glasses sit tilted or low on your nose, the camera's field of view will be angled downward and detection accuracy for objects at standing height will suffer.

2.3 Powering On

1. Make sure the glasses are charged (see Section 2.5).
2. Power on the glasses with a brief press of the capture button. From deep sleep the firmware wakes on any button event; you do not need to hold the button down.
3. Open the Finding the Way application on your smartphone and connect to the glasses (see Part 3). The glasses appear as “ESP32-CAM” in the Bluetooth scan list.

NOTE. The smart-glasses firmware is power-managed for a 6–10 hour battery life. As a fallback, the firmware also takes a passive burst on a 30-second interval if no other catalyst triggers.

2.4 Understanding Visual Detection

Once both peripherals are connected, the smart glasses operate automatically. You do not need to press any buttons during normal use.

1. The chest pack detects an object within range and notifies the application over BLE.
2. The application forwards a single-capture command to the smart glasses (rate-limited to once every 3 seconds while an object remains present).
3. The glasses capture a single frame and transmits it to the smartphone.
4. The smartphone runs an ONNX YOLO model trained on Open Images v7. It identifies objects, estimates each one's distance in feet, and classifies motion as approaching, moving away, crossing left, or crossing right.
5. The smartphone announces the results aloud. Objects within five feet are spoken first, regardless of what else is in the scene.

Example announcements you may hear:

- “Warning: a person approaching, about three feet away.”
- “Chair ahead, about five feet away.”
- “Two people crossing right, about eight feet away.”
- “Car moving away, about twelve feet.”

NOTE. The visual system identifies objects from approximately 600 categories defined by the Open Images v7 label set. Items outside that label set will not be announced. Common mobility hazards such as curbs, puddles, and small steps may not be reliably detected in the current version.

TIP. You can trigger a burst capture by pressing the capture button on the glasses. This is useful for quickly identifying what is ahead of you without waiting for the ultrasonic system to fire.

2.5 Charging the Smart Glasses

1. Connect the USB-C cable to the charging port on the glasses frame.
2. Connect the other end to a power source.

3. The on-board LED will flash red while charging. Full charge from empty takes approximately 1 to 1.5 hours.
4. Battery life is approximately 6–7 hours under heavy continuous capture, 8–10 hours under typical mixed use, and 12–15 hours when mostly idle.

⚠ WARNING. Do not charge the glasses while wearing them. Do not bend or compress the frame arms forcefully, the integrated electronics can be damaged by physical stress to the frame.

2.6 Caring for the Smart Glasses

- Clean the camera lens with a dry microfiber cloth only. Do not use liquid cleaners, paper towels, or abrasive materials. Scratches will degrade image quality and detection accuracy.
- Do not wear the glasses in rain, heavy humidity, or near water. The glasses are not waterproofed.
- Store the glasses in a protective case when not in use. Avoid leaving them in direct sunlight or in hot enclosed spaces such as a parked car.
- Do not attempt to disassemble the frame. The MCU board, battery, and camera are integrated components.
- If the glasses fail to connect or stop transmitting images, power them off and on. If the problem persists, see Troubleshooting.

Part 3: Mobile Application

The Finding the Way application runs on your Android smartphone and serves as the central hub connecting the chest pack and the smart glasses. All detection announcements, system controls, battery monitoring, and settings are managed through the application.

3.1 Installing the Application

1. Verify your Android device is running Android 7.0 (Nougat, API 24) or higher.
2. Make sure Bluetooth is enabled.
3. Install the Finding the Way APK the download link on Github. Installing from a downloaded file will require you to allow installation from unknown sources if not already enabled.
4. When prompted, grant the following permissions: Bluetooth Connect and Bluetooth Scan, Location, and Notifications. The app cannot maintain background BLE connections without Notifications and the Foreground Service subsisting in the notifications bar.

NOTE. The application requests no microphone access. Voice output is produced through Android's built-in TextToSpeech engine, which does not require microphone permission. The app is designed for TalkBack compatibility; enable TalkBack before launching the app for the first time so that all controls are correctly announced.

3.2 Connecting to the Devices

1. Make sure the chest pack is powered on and the smart glasses are awake.
2. Open the Finding the Way app. The main page appears with five large buttons.

3. Tap Connect Devices, then tap Scan. A list of nearby BLE devices populates. The chest pack appears as “Navigation Vest” and the smart glasses appear as “ESP32-CAM”. If the devices are not found, tap Scan to initiate a new scan.
4. For each discovered device, tap the row, then tap the role you want to assign (Smart Glasses or Navigation Vest). The app uses a confirm-on-second-tap pattern: the first tap announces the action and the second tap performs it.
5. Once both connections succeed, the connection banner on the main page reads “Glasses and vest connected”.

NOTE. If the chest pack does not appear in the device list, make sure it is powered on and within about 10 metres. Tap Scan again to repeat the discovery (each scan runs for about 16 seconds). If the glasses do not appear, press the capture button briefly to wake them from deep sleep, then rescan.

TIP. You can connect the chest pack and glasses in either order. The application maintains both BLE connections through the Assistive Runtime foreground service, and it will automatically retry either link if the connection drops.

3.3 Navigating the Application

The main page provides five large buttons:

Button	Function
Connect Devices	Opens the BLE scan and pairing page for the smart glasses and the chest pack.
Device Status	Shows connection state and battery level for both peripherals. Provides buttons for Single and Burst Capture.
Settings	Provides system toggles, runtime modes, and the volume control.
Saved Images	Displays the JPEG frames captured by the smart glasses.
Activity Log	A timestamped technical log of session events, useful for troubleshooting.

3.4 Adjusting Settings

From the main page, tap Settings. The Settings page contains the following controls:

- **System Power:** master switch for the entire system. Turning this off forces both ultrasonics and haptics off on the chest pack.
- **Haptic Feedback:** enables or disables the chest-pack vibration motors.
- **Ultrasonic Sensors:** enables or disables the chest-pack distance sensors.
- **Audio Feedback:** enables or disables spoken announcements through the phone's TextToSpeech engine.
- **No Objects Announcement:** when on, the app says “No objects detected” after a capture cycle that finds nothing. When off, silence indicates no detections.
- **Assistive Runtime Mode:** a radio group with three choices — Assistive Active (BLE + detection + speech), Assistive Silent (BLE + detection, speech muted), and Phone Priority (assistive speech

paused immediately). Switching from Active to either of the others stops any in-flight announcement.

- **Background Runtime Safeguard:** a long-press confirmation control for disabling the background BLE runtime. This prevents accidental shutdown of the foreground service.
- **Glasses Picture Taking:** gates whether the smart glasses are allowed to capture and stream new frames while BLE remains connected.
- **Volume Control:** a slider that sets the announcement volume in addition to the phone's hardware media volume buttons.

NOTE. If you turn off Ultrasonic Sensors while the smart glasses are not active, the chest-pack firmware will challenge the request through the application (“Are Glasses On?”) and, if the answer is no, override the change to keep the vest active and announce a safety override.

TIP. In noisy environments, switch to Assistive Silent and rely on the haptic vest. In a quiet environment where you prefer spoken identification only, you can disable haptic feedback and rely on audio.

3.5 Understanding Audio Announcements

The application announces detected objects through Android's TextToSpeech engine. Speech is prioritised so the most urgent information is spoken first:

Priority	Trigger	Example announcement
URGENT	Any object within 5 feet	“Warning: a person, about three feet away.”
HIGH	Moving object at medium range (5–15 ft)	“Car moving left, about ten feet away.”
MODERATE	Stationary object nearby	“Chair ahead, about six feet.”
LOW	Distant or retreating object	“Bench, about twenty feet.”

A maximum of seven items are announced per capture cycle to prevent information overload in crowded scenes. Items of the same class, motion state, and approximate distance are grouped: for example, “three people approaching, about four feet away.”

NOTE. Low-battery announcements interrupt other speech. The chest-pack firmware sends battery messages every five seconds and dedicated BATTERY_LOW or BATTERY_CRITICAL messages at the 25%, 10%, and 5% thresholds. Recharge the chest pack at your earliest opportunity to avoid an unexpected shutdown.

3.6 Volume and Background Behaviour

- Volume can be adjusted with the Volume Control slider in Settings or with the phone's hardware volume buttons during use.

- If the app is moved to the background, the AssistiveRuntimeService keeps the BLE connections alive and continues delivering alerts via a foreground notification. Announcements will pause during phone calls and resume automatically afterwards.

3.7 Caring for Your Smartphone

- Keep the smartphone charged. The application and the BLE radio draw additional battery. For sessions longer than two hours, start with a full charge or carry a portable charger.
- Do not cover the phone's speaker during use.
- If you use earphones or a Bluetooth speaker, the TTS audio routes there automatically through Android's standard audio routing.

Troubleshooting

Problem	Likely cause	Solution
Chest pack does not power on	Battery pack depleted or not powered on	Charge the pack for at least 30 minutes.
App does not find chest pack	Chest pack is off, out of range, or already connected to another device	Verify the chest pack is powered. Move within 5 meters. If necessary, restart the chest pack by powering off and on.
App does not find smart glasses	Glasses are in deep sleep, off, or battery is depleted	Press the capture button briefly to wake the glasses; if no response, charge for at least 30 minutes.
No haptic feedback despite obstacles	Ultrasonic Sensors or Haptic Feedback toggled off, BLE connection lost, or System Power off	Confirm physical toggle switches are toggled on. If connected to mobile application, on the Settings page, confirm System Power, Haptic Feedback, and Ultrasonic Sensors are all on. On Device Status, confirm the chest pack is connected.
No spoken announcements	Audio Feedback toggled off, runtime mode set to Assistive Silent or Phone Priority, phone media volume at zero, or BLE link to glasses lost	On the Settings page, check Audio Feedback and Assistive Runtime Mode. Increase phone media volume. On Device Status, check the glasses connection.
Haptic feedback fires constantly with no objects present	Sensor interference from a reflective surface (glass walls, metal shelving) or an obstruction touching the sensor face	Move away from the surface and clear any clothing or accessories from the sensor faces. If the issue persists, toggle Ultrasonic Sensors off and back on.

Problem	Likely cause	Solution
App crashes or freezes	Memory pressure from other apps	Close other background apps; restart the application. If the problem persists, restart the phone.
BLE connection drops repeatedly	Phone too far from the peripherals or RF interference present	Keep the phone within about 10 meters of both devices. Avoid areas with dense Wi-Fi or other 2.4 GHz congestion.
Safety interlock blocks a settings toggle	The change would leave the system without an active sensing modality	Enable at least one sensing modality before disabling the other. The application will announce the reason for the block.
Low-battery alert recurs shortly after charging	Pack capacity has degraded over many charge cycles	Contact support for battery replacement guidance. Li-Ion packs typically support 300–500 full charge cycles.

Quick Reference Card

Print this page or save it as a screenshot for use in the field.

Startup Sequence

1. Charge all three devices fully before your first session.
2. Put on the chest pack and adjust the straps.
3. Power on the battery pack.
4. Put on the smart glasses.
5. Wake the smart glasses with a brief press of the capture button.
6. Open the Finding the Way app on your phone.
7. Tap Connect Devices and connect to “Navigation Vest” and “ESP32-CAM”.
8. Confirm both devices show as connected on the main page.
9. Begin navigation.

Shutdown Sequence

1. Press the red push button on the inside of the chest pack once to power off the chest pack.
2. Long-press the smart-glasses capture button (about 2 seconds) to put the glasses into deep sleep.
3. Remove the glasses, then remove the chest pack.
4. Store all components in a cool, dry place.

Physical Button Reference

Control	Location	Action
Power button	Chest Pack Internals	Push red button once to power on and once to power off
Switch 1 (Haptics)	Right side of housing unit	Position toggles haptic feedback on or off (subject to app override when connected).
Switch 2 (Ultrasonics)	Left side of housing unit	Position toggles ultrasonic sensing on or off (subject to app override when connected).
Capture / power button	Smart-glasses frame	Brief press causes burst capture when connected to mobile app and wakes from deep sleep otherwise. A 2s hold causes the glasses to enter deep sleep.

Haptic Distance Reference

Pulse pattern (on / off)	Distance and meaning
100 ms / 100 ms (very rapid)	2 ft — stop or turn immediately
100 ms / 300 ms (fast)	3 ft
100 ms / 500 ms (medium)	4 ft
100 ms / 700 ms (slow)	5 ft
100 ms / 900 ms (very slow)	6 ft — awareness only
No vibration	Beyond 6 ft, or no object detected in this zone

Contact and Support

For technical support, replacement parts, or feedback on the system, please contact:

The Citadel, Department of Electrical and Computer Engineering
Reference: 2026 Capstone, Team Name: Circuit Syndicate

For battery recycling guidance, visit call2recycle.org or contact your local municipal recycling authority.

NOTE. Finding the Way is a research prototype developed as part of the ELEC 421/422 capstone design sequence at The Citadel. It has not received FDA clearance as a medical device. Use it as a supplementary navigation aid only.